

John Isaac Aziz

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SUMMARY

Senior Software Engineer with 3+ years of experience delivering full-stack and cloud-native solutions using Python. Currently building a new AI powered EdTech platform at Lyah, with prior experience building distributed systems, developer tools, and internal packages at startups and scale-ups.

Recognized as a former Microsoft MVP for impactful contributions to the tech community, including open-source tools, educational content, and developer advocacy. Proven ability to lead technical initiatives, automate workflows, and deliver scalable solutions across cloud platforms.

Recent PROFESSIONAL EXPERIENCE

Lyah Ed Tech

Dec 2025 – Present

Senior AI Software Engineer (Python, Vue.js)

- Architected and built from scratch the multi-agent framework powering all of Lyah's AI features: a modular, node-based LangGraph system in Python (replacing legacy n8n) with workflow logging, Google Chat notifications, Celery orchestration, a provider-agnostic multi-LLM layer with automatic fallback (OpenAI/Anthropic/Google), and Opik observability.
- Built Lyah's core course and lecture generation experience end to end: AI agents that draft courses/lectures from a prompt, a YouTube video, or documents, plus the Session Studio and Course Builder authoring UI in Vue 3 with an AI proposal-review flow and client-side diffing.
- Built a full-stack AI tutor/copilot assistant: a conversational agent with intent classification and document-grounded retrieval (RAG), surfaced through a real-time WebSocket chat with persistence, inline editing/feedback, and export-to-Word.
- Developed an LLM evaluation and automated prompt-optimization pipeline (LiteLLM + Opik MetaPrompt), reaching 96.3% decision accuracy with precision/recall/F1 metrics and production drift monitoring.
- Delivered full-stack instructor analytics combining computed metrics with LLM-generated narrative commentary into viewable, downloadable reports.
- Integrated external LMS platforms (Moodle, Canvas) via the LTI standard, and established the testing/CI/CD foundation (pytest coverage gating, uv, AI-assisted PR workflows).

Orange Business

Nov 2024 – Dec 2025

Software and Data Engineer (Python, React.js)

- Collaborate with business stakeholders to gather requirements, break down features, and deliver backend and frontend components using FastAPI, Python, and React.js.
- Build and deploy automation scripts and services with Kubernetes CronJobs, Helm, and GitLab CI/CD.
- Developed and published an internal Python package for shared logic, integrated with CI/CD pipelines and Docker images for reuse across applications.
- Support, maintain, and lead development on 3 applications, including ownership of one as sole developer and managing their migration to a new cloud environment.
- Integrated systems with OIDC (Orange Connect) and added Redis caching to improve API performance.
- Built RAG pipelines and automated document ingestion workflows for contracts and SharePoint using pgvector, OpenAI, and scheduled batch jobs.
- Provide technical leadership and code reviews offering guidance and direction across projects for my team.
- Support production deployments and troubleshoot live issues to ensure system reliability.

Fabrica Dev

Aug 2024 - Nov 2024

Software Engineer (Python, Vue.js)

- Design, develop, and implement backend services using Python, MySQL, MongoDB, Vue.js.
- Integrate software components into a fully functional software system.
- Collaborate with cross-functional teams to define, design, and ship new features.
- Created a software Integration with Egyptian Tax Authority to generate and submit digital invoices.

GEMINDZ

Mar 2024 - Aug 2024

Software Developer (Python, Vue.js)

- Designed, Implemented and Deployed a Full stack web application using Python (FastAPI), Vue.js, and Azure.
- Implemented multiple OS independent monitoring utilities and integrated them with a distributed software.
- Automated on Device Testing using Pytest, GitHub and Gitlab Actions.
- Built a Python package to manage multi-state hardware devices and automate firmware flashing.

OSS Contributor

Jan 2021 - Present

- Maintainer of Pallets Ecosystem Python [CacheLib](#) package.
- Developed the landing website for Azure SQL, dramatically boosting resource accessibility and streamlining navigation for a diverse audience, attracting over **5k monthly** views.
- Authored over +12 technical blogs, [3 Microsoft Learn Modules](#), and one Microsoft Learn tutorial attracting over **150k** views combined, which significantly improved learner engagement on the Microsoft Learn platform.
- Engineered a Python package for **markdown link validation** that is utilized in key Microsoft GitHub repositories, achieving over **24k monthly** downloads to ensure content integrity across multiple platforms.
- Automated link validation in [Microsoft's Generative AI repository using GitHub Actions](#), enhancing the learning experience for over **100k** learners by ensuring high-quality, error-free educational content.
- [Featured in Microsoft's Contributor Stories](#) blog series for my impactful educational contributions, reaching a global audience and enhancing Microsoft's community-driven educational efforts.
- [Implemented 2 features and fixed 1 bug](#) for the Azure Developer CLI (azd) using Go Language.

EDUCATION

Modern Academy for Engineering and Technology

Sep 2018 - Aug 2023

B.Sc. in Computer Engineering and Information Technology (CGPA: 3.44/4.0)

- Thesis (Final Year Project): Centralized and Automated Medical Image Analysis Website Using Deep Learning. (Grade: A+ Excellent)
- Ranked 4th in my class in a cohort of over 150 students.

AWARDS/SCHOLARSHIP RECORDS

[Microsoft MVP](#)

Jan 2024

[Top MENA GDSC Lead 2020-2021](#)

Aug 2021

PROFESSIONAL CERTIFICATIONS

Expert/Associate

[Microsoft Certified: Azure AI Engineer Associate](#)

Mar 2023 - Mar 2027

[Google Cloud Certified: Professional Cloud Architect](#)

Feb 2024 - Feb 2028

[AWS Certified Cloud Practitioner](#)

Feb 2023 - Dec 2028

[Google Cloud Certified: Associate Cloud Engineer](#)

May 2022 - May 2028

Fundamental

[Microsoft Certified: Azure Data Fundamentals](#)

Oct 2023

[Microsoft Certified: Power Platform Fundamentals](#)

Oct 2023

[Microsoft Certified: Azure Fundamentals](#)

Aug 2022

[Microsoft Certified: Azure AI Fundamentals](#)

Jun 2022

PUBLICATIONS

Microsoft Learn

[Introduction to large language models](#)

Dec 4, 2023

Microsoft Tech Community

[Bring your own Data to Azure OpenAI: Step-by-Step Guide](#)

Sep 7, 2023

[Deploying a Large Language Model \(GPT-2\) on Azure Using Power Automate: Step-by-Step Guide](#)

Jun 21, 2023

VOLUNTEERING EXPERIENCE

Leader – Azure Egypt User Group

Feb 2021 - Jun 2025

Organizer – Google Developer Groups Cloud Egypt

Feb 2021 - Feb 2025

Mentor – Imagine Cup

Dec 2023 - Feb 2023

Judge – Imagine Cup Junior Competition

May 2023

Community Leader – Google Developer Student Clubs

Aug 2020 - Jul 2021

Vice President – Enactus Modern Academy

Aug 2019 - Aug 2020

LANGUAGES

Arabic (Native), English (Fluent), French (Beginner)

INTERESTS

Research Interest

Open Source, Community Advocacy, Cloud Development, Full Stack Development.

Other Interest

Volunteering, listening to music, watching tv series & movies, playing musical instruments.